

Legged Robots: Project 1

30.09.2025

Plan

Fundamentals (ungraded)

- **W1-3** (09.09, 16.09, 23.09)
 - Introduction + double pendulum kinematics and dynamics
 - Jacobian (Cartesian PD + Force Control)
 - Inverse Kinematics (compare with force control)
 - Single-leg hopping

Mini-Project 1

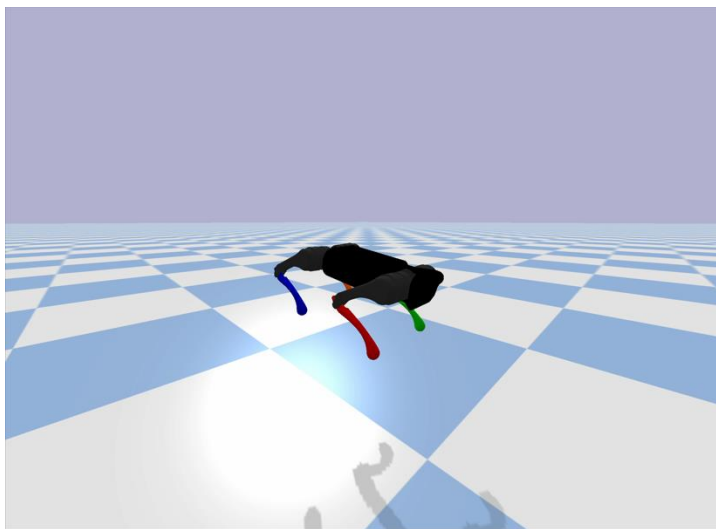
- **W4-6** (30.09, 07.10, 14.10)
 - Model-based control of a Quadruped
[MP1 Report – 15% of grade]

Mini-Project 2

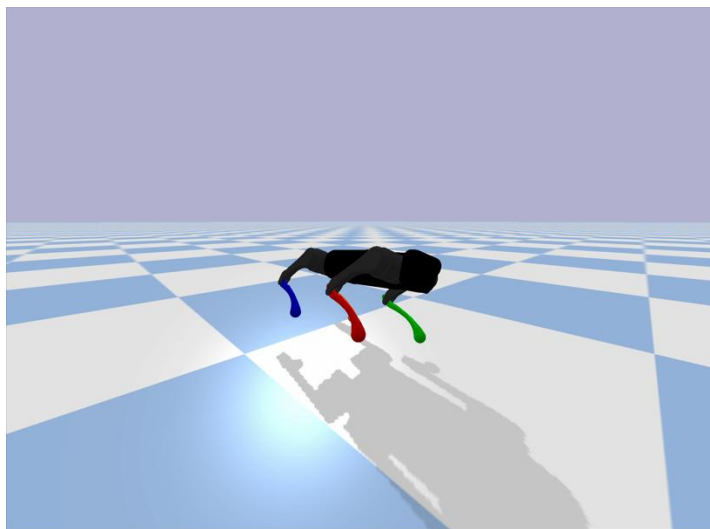
- **W7-14** (28.10-16.12)
 - Quadruped CPG Trot
 - Quadruped Locomotion Project (CPGs, Deep RL)
[MP2 Report – 35% of grade]

Quadruped Jumping

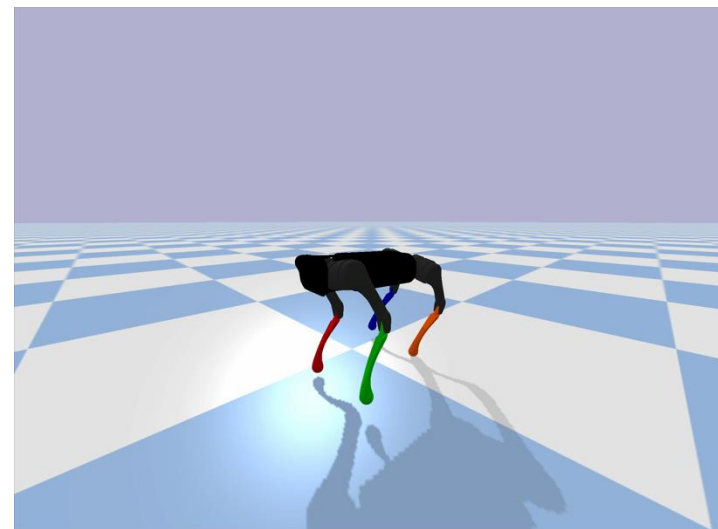
Forward jump



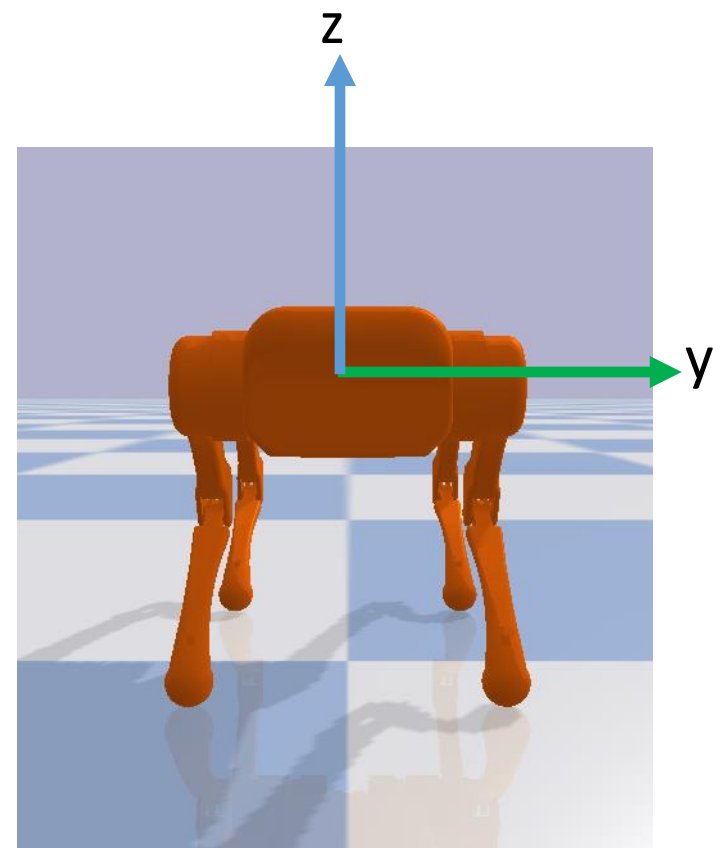
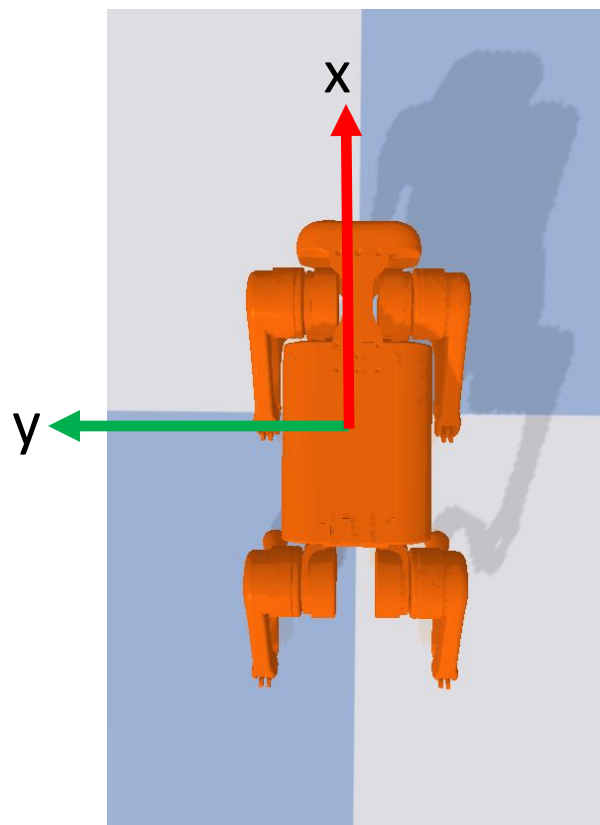
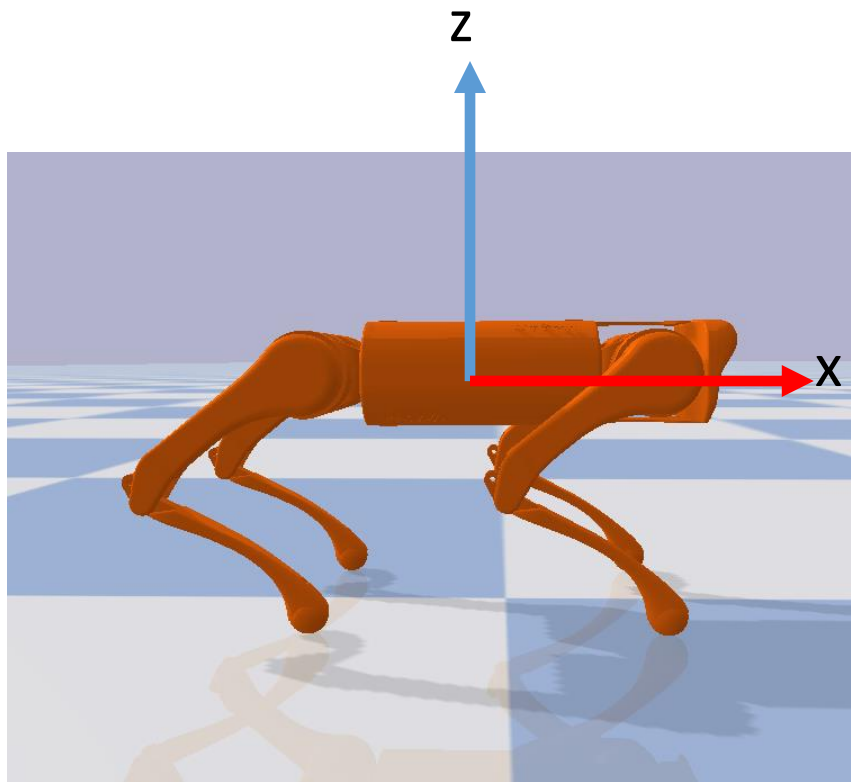
Lateral jump



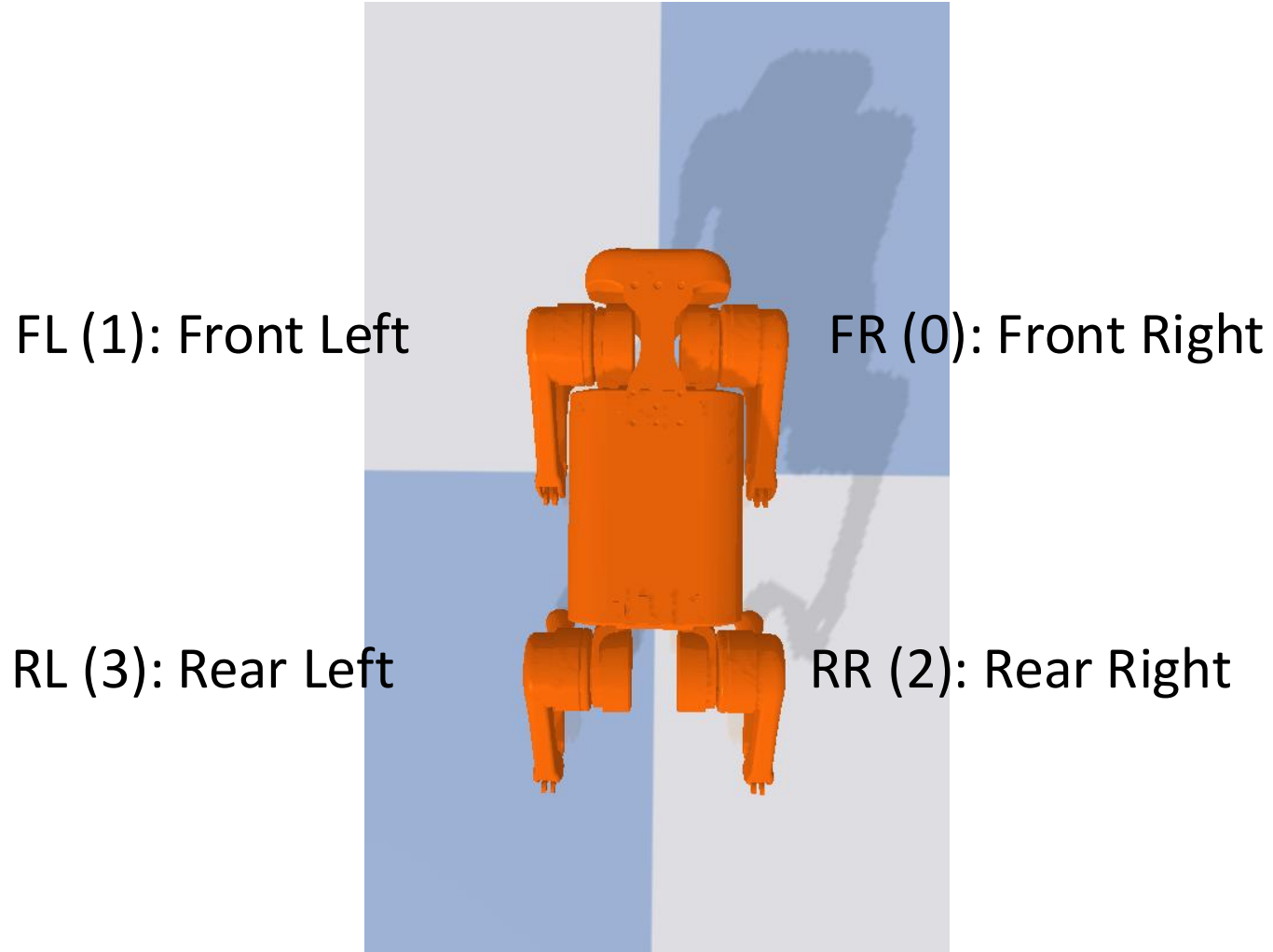
Twist jump



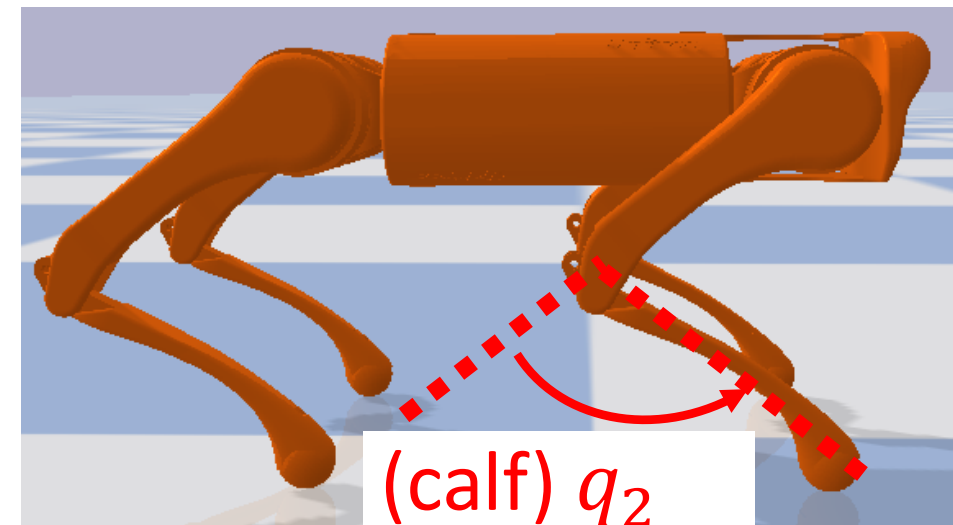
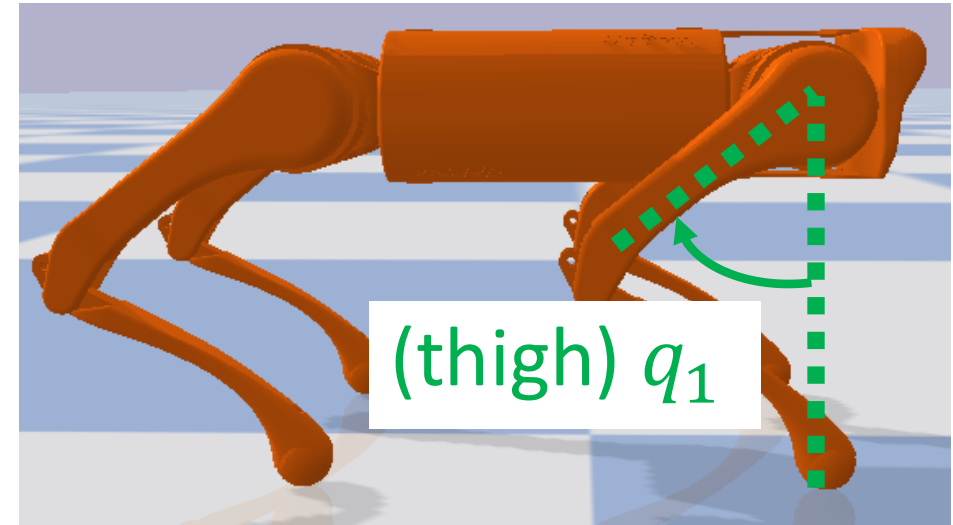
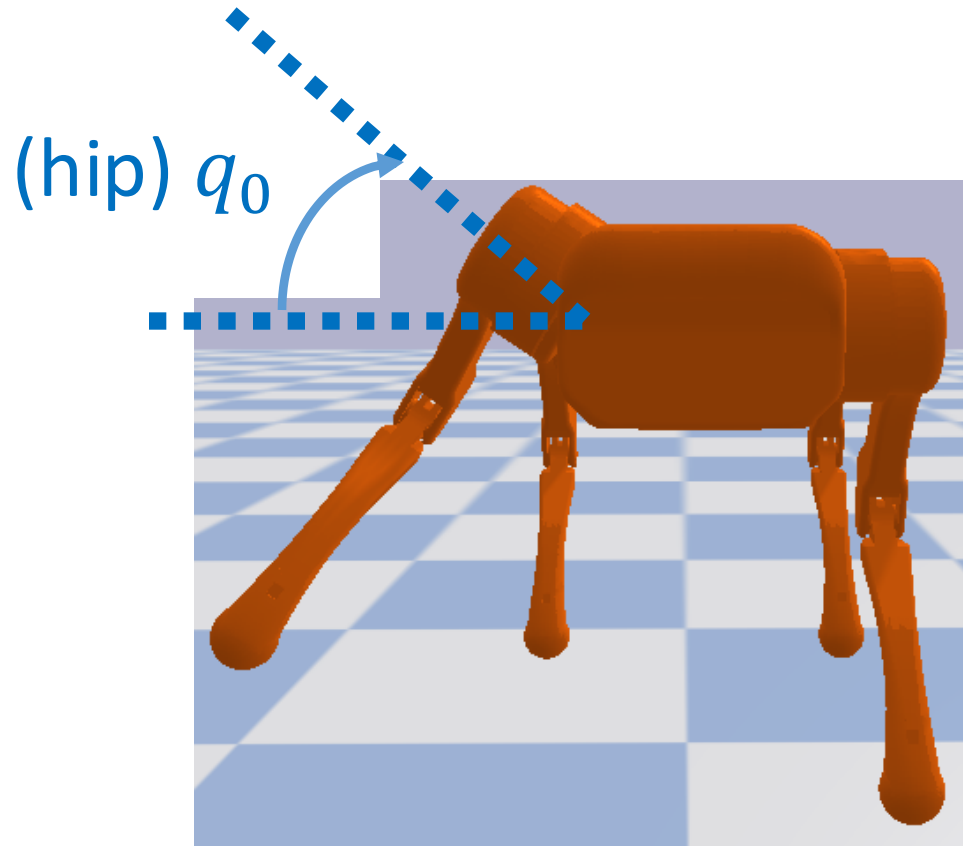
Quadruped Model Reference Frame



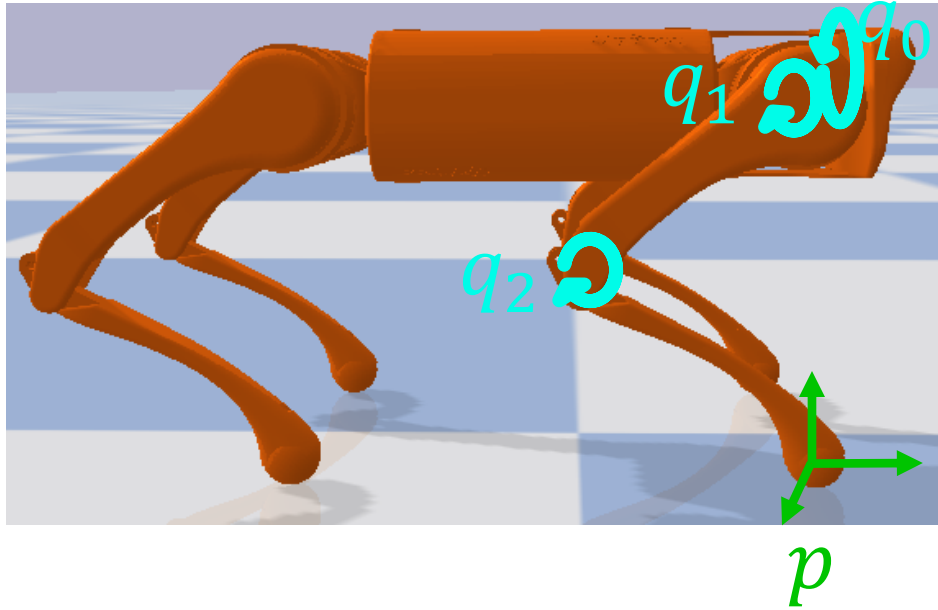
Quadruped Model Leg References



Quadruped Model Joint References



Joint angles $\leftrightarrow$ Cartesian space (leg frame control)



$$p = f(q) \quad \text{Forward kinematics}$$

$$q = f^{-1}(p) \quad \text{Inverse kinematics}$$

$$\dot{p} = v = J(q)\dot{q} \quad \text{Foot linear velocity}$$

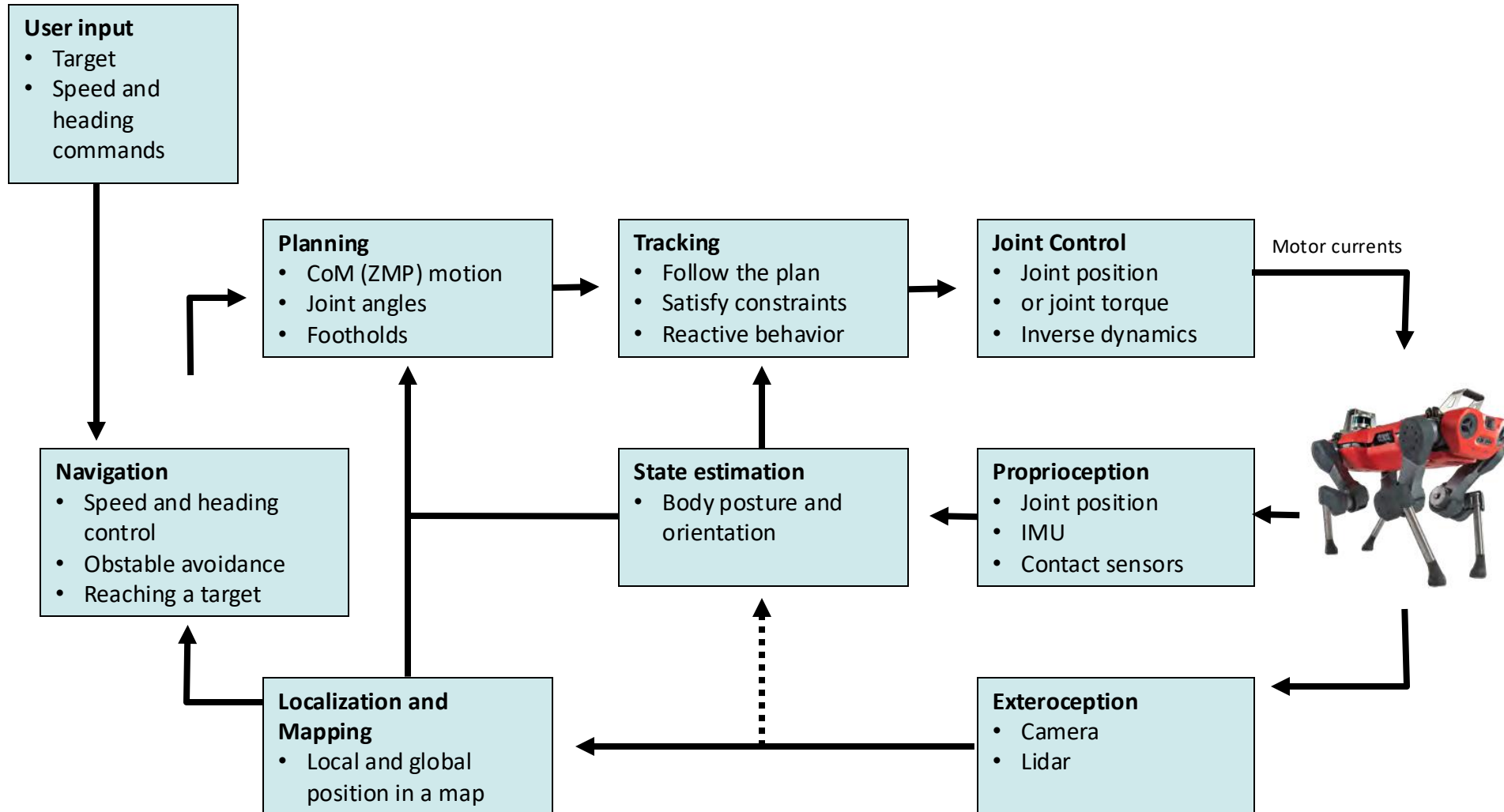
$$\tau = J^T(q)F \quad \text{Map desired end effector force to torques}$$

$$\tau_{joint} = K_{p,joint}(q_d - q) + K_{d,joint}(\dot{q}_d - \dot{q}) \quad \text{Joint PD}$$

$$\tau_{Cartesian} = J^T(q)[K_{p,Cartesian}(p_d - p) + K_{d,Cartesian}(v_d - v)] \quad \text{Cartesian PD}$$

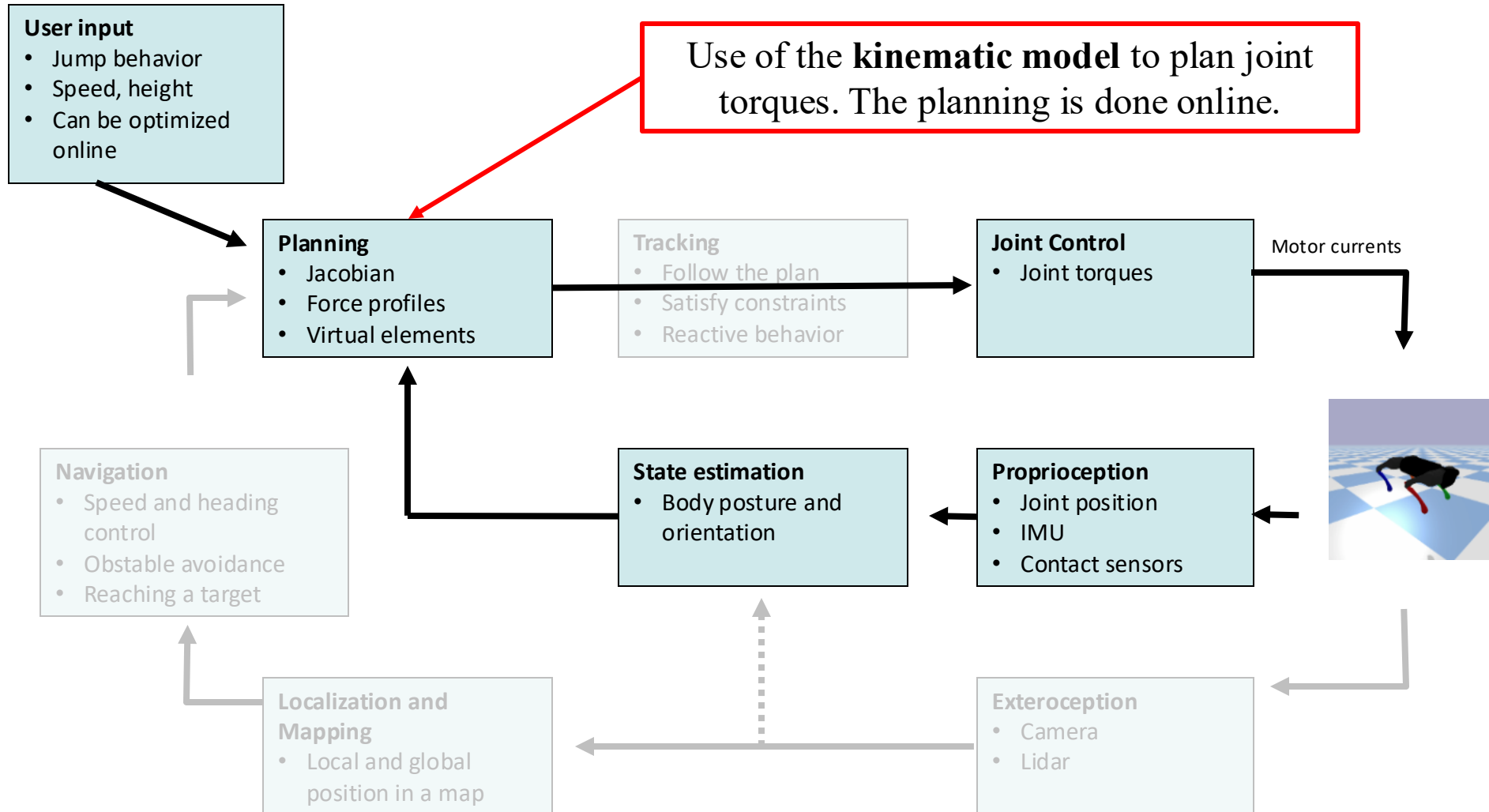
Typical control architecture

Lecture 3



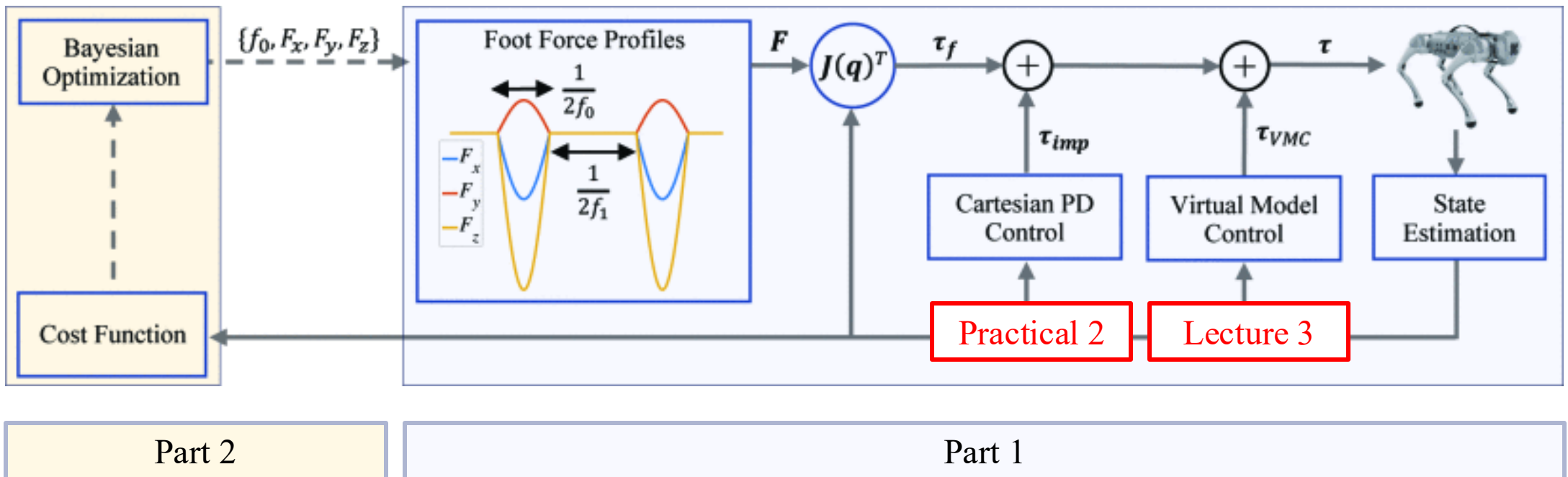
Typical control architecture

Lecture 3



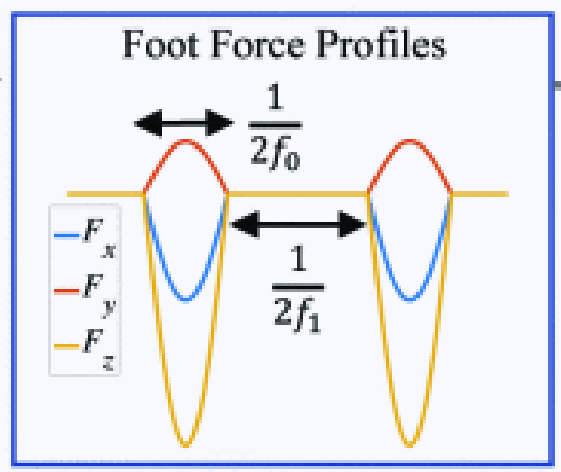
Jumping Controller

Practical 4



G. Bellegarda, M. Shafiee, M. E. Özberk, and A. Ijspeert, "Quadruped-Frog: Rapid Online Optimization of Continuous Quadruped Jumping," in 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024, pp. 1443–1450.

Force Profiles

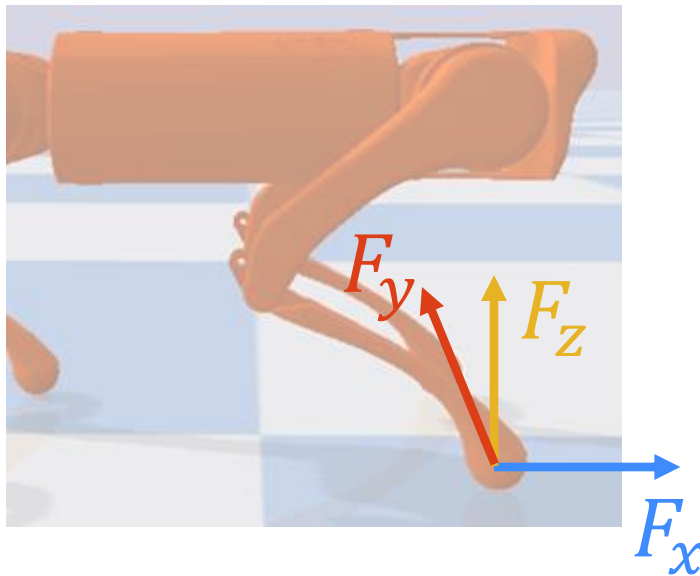


Single oscillator with 2 frequencies:

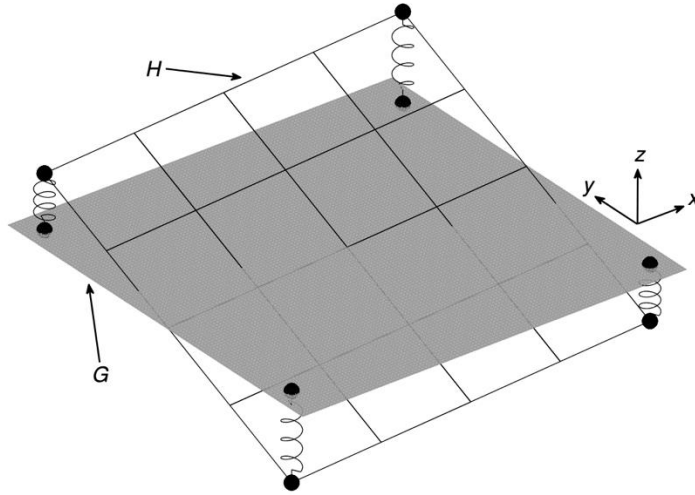
$$\dot{\theta} = 2\pi f_i, \quad \text{where} \quad f_i = \begin{cases} f_0 & \text{if } \pi \leq \theta < 2\pi, \\ f_1 & \text{if } 0 \leq \theta < \pi \end{cases} \quad \begin{matrix} \text{(impulse)} \\ \text{(idle)} \end{matrix}$$

Force scaled by oscillator phase:

$$\mathbf{F}_i = \begin{cases} \begin{bmatrix} F_x & F_y & F_z \end{bmatrix}^\top \sin(\theta) & \text{if } \sin(\theta) < 0 \\ \mathbf{0}_{3 \times 1} & \text{otherwise} \end{cases}$$



Virtual Model Control

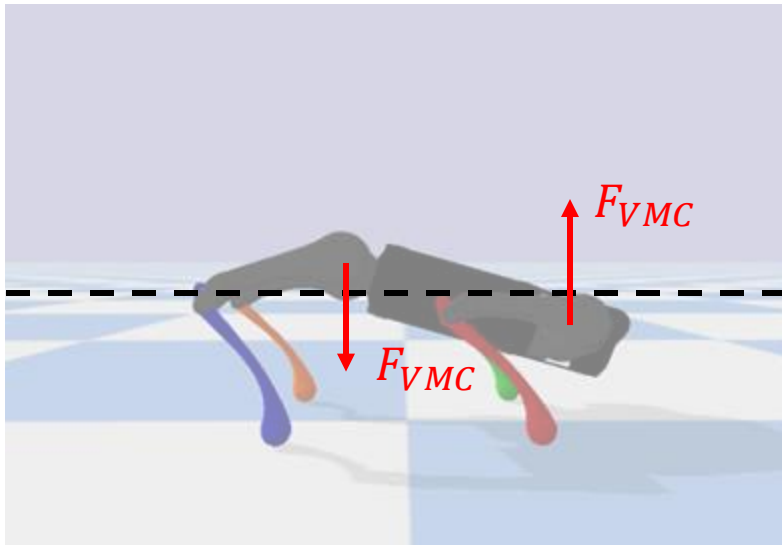


Obtain robot's base plane in world coordinates:

$$P = R \begin{bmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

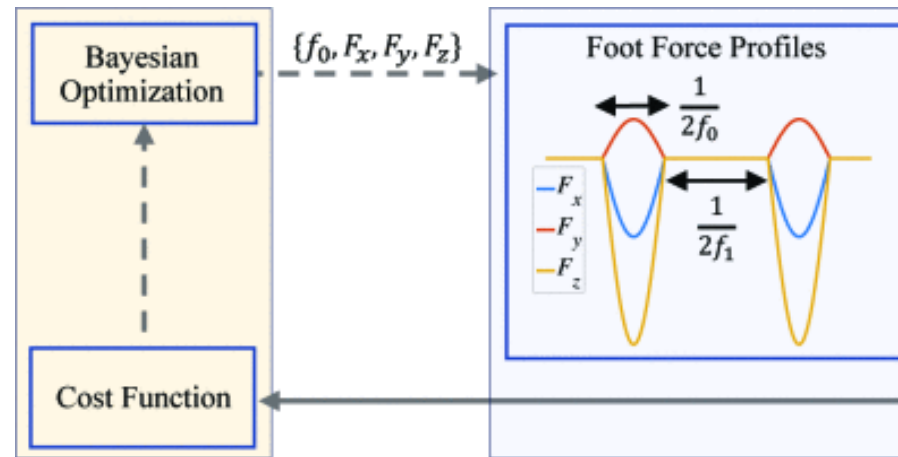
Attach virtual springs in corners:

$$F_{VMC} = \begin{bmatrix} \mathbf{0}_{2 \times 4} \\ k_{VMC} ([0 \quad 0 \quad 1] P) \end{bmatrix}$$



M. Ajallooeian, S. Pouya, A. Sproewitz, and A. J. Ijspeert, "Central Pattern Generators augmented with virtual model control for quadruped rough terrain locomotion," in 2013 IEEE International Conference on Robotics and Automation, 2013, pp. 3321–3328.

Optimization



- Optimize parameters of foot force profiles (or other parameters?)
- Design methodology (variables and costs) for the 3 types of jumping + continuous hopping
- Script available to optimize your implemented controllers using Optuna

<https://optuna.readthedocs.io/>

Assignment

- Groups of 3 students (register on Moodle!)
- Project instructions are on Moodle
- Submit 1 zip file per group **before Tue 21st Oct 23:59** (no practical!)
- Download code at <https://gitlab.epfl.ch/lgevers/lr-miniproject-1>

Generative AI Policy

- Use of generative AI for the projects is allowed (but optional)
- Explicit declaration about AI use in your report (helpful insights!)
 - How you prompt it
 - Where it's been helpful
 - Where it hasn't been
- When using AI for coding: modify the individual functions (not the full codebase)
 - TA support might otherwise be limited